ECO203: Introductory Econometrics

Empirical Project – Final Report

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Introduction and Motivation

The Shiv Nadar University Central Library is the soul of the institution and it has started acquiring, processing, preserving and disseminating information to the user community with the motto of empowering knowledge. It has an extensive collection of over 40,000 books including scientific and technical journals and electronic reference materials for satisfying the academic and research needs of students as well as the faculty community. Despite the fact that the numbers of books in the library are so many, it is logistically impossible to provide each student with a book that they might require for a particular course. We are essentially trying to find out "To what degree is the number of books issued in a semester in Shiv Nadar University affected by the number of library cards in possession, individual preference of books over soft copies and the number of days, on average, a person is late in returning an issued book"? We came up with this question after being a victim of this issue. Time and again, we faced a problem in being unable to issue a reference book either due to unavailability or lesser number of copies or something else. To find out what the root cause of this issue could be, we have taken this up.

Since most of our respondents have a strict preference of books over soft copies (our survey said: 75% of the respondents prefer hard copies to soft copies, however, only 10% of them get to actually use hard copies) we would like to know the factors that make or prevent them from actually issuing books. These factors that prevent them could range from shortage of books in the library to sheer laziness on the part of the students. The inability of the library to supply expensive books could also be a reason. Since this project is SNU specific, there is no previous literature on the same.

Data

1. **Source of Data:**

   We have used primary sources of data that were be obtained by using a survey (Refer Appendix 1). This is partly due to the unavailability of secondary data and partly due to the higher accuracy of primary data that provide first-hand testimonials.

2. **Target Population:**

   Our study is restricted to the students studying in Shiv Nadar University. The target population is then the undergraduate students in their 1st, 2nd, 3rd and 4th years. We have collected the responses of 85 students belonging to this population. Since the population is spread across all streams and all batches, there is no scope of bias.

3. **Dependent Variable:**

   a. Books issued (booksissued):

      This variable denoted the number of books that are issued in a semester.

4. **Independent Variable:**

   a. Library Cards (libcards):

      This variable measures the number of library cards a student has. At the beginning of the first semester, each student is given 5 library cards for the purpose of issuing books. Over semesters, some students tend to lose their cards because of their carelessness or because they are stolen. Due to this reason we observed a variation from 0-5 for this variable.

b. Late Returns (latereturn):

This variable measure the number of days by which the return of a book is delayed. Many a times, either due to exams or due to a deficiency in the number of copies of a particular course book in the library, students hoard on books by not returning it on time. By doing this, they are able to avail for the book throughout the semester without worrying about its availability in the library. This effects the number of books issued, as other students are not able to avail for a book, which should ideally be present in the library.

c. Preference of Book (pref_book):

For any course that a student takes, he/she has an option to either study from a hard copy or a soft copy of the prescribed reference book. Based on this, we have introduced a dummy variable- preference of the form of the book (pref_book), which takes binary values-- the value 1 if the student prefers a hard copy and 0 if the student prefers a soft copy.

d. Error Term:

1. The error term ("u") contains the class strength of courses, number of exams for a particular course, difficulty level of a course, course structure etc.

Methodology

Using the data we received, we carried out a multiple regression analysis using a linear probability model.

The regression equation is given below:

$$booksissued = \beta 0 + \beta 1\ (libcards) + \beta 2\ (latereturn) + \beta 3\ (pref_books) + u$$

We ran our regression model using the method of OLS [Ordinary Least Squares] to get estimates of the parameters βj [where $j \in (1, 3)$]. Running a multiple regression on Stata will allow us to estimate the effect of various explanatory variables on our qualitative dependent variable i.e. booksissued.

Scope of Error

1. Subjective questions: The questions asked in the survey are very subjective in the sense that, their answers may vary from person to person depending on their preferences.
2. Small sample size: Our sample size is only 85. Meaning, the variation in our data would be very minimal.
3. Fewer independent variables: Had we kept more independent variables, they would've explained our results even better.
4. Possible existence of positive correlation between independent variables and the error term, which could cause bias.

Summary of Variables

Here, we have the Mean and Standard Deviation values for our independent variable and each of our dependent variables.

```
. summarize booksissued latereturn pref_books libcards
```

Variable	Obs	Mean	Std. Dev.	Min	Max
booksissued	85	2.670588	2.112256	0	10
latereturn	85	7.2	15.14753	0	120
pref_books	85	.7529412	.4338609	0	1
libcards	85	3.270588	1.348617	0	5

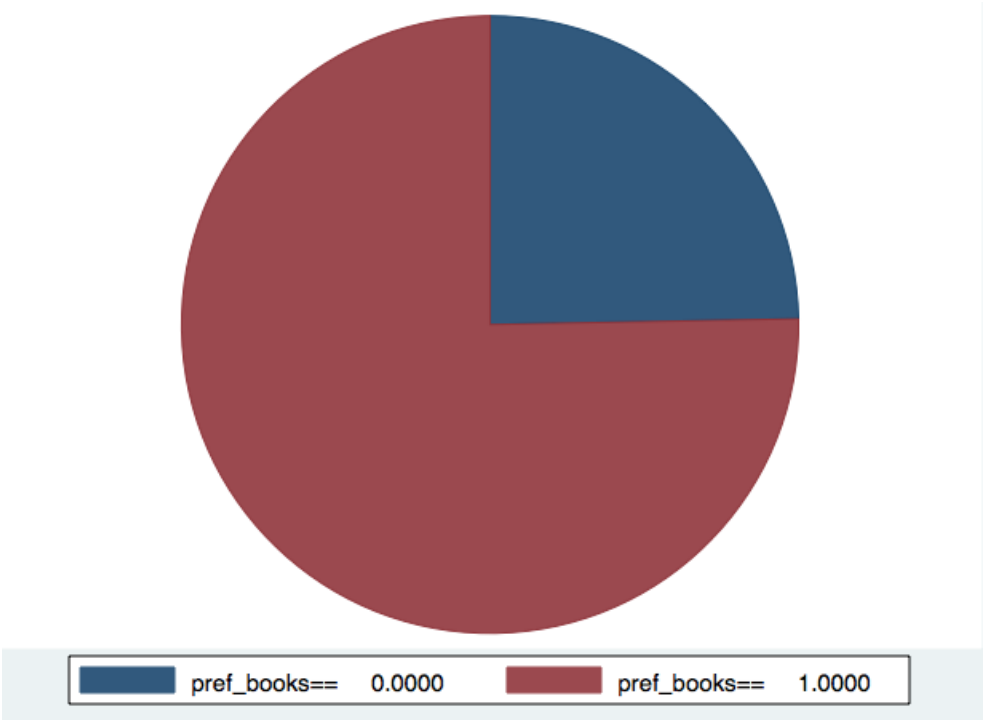
Dummy Variable

Out of the 85 observations in our data, we observed that 21 students preferred a soft copy over a hard copy (pref_books took a value of 0).

64 students preferred a hard copy over a soft copy (pref_books took a value of 1).

```
. tabulate pref_books, sum(booksissued)
```

pref_books	Summary of booksissued Mean	Std. Dev.	Freq.
0	2.1904762	2.5810666	21
1	2.828125	1.9318113	64
Total	2.6705882	2.1122558	85



pref_books	Freq.	Percent	Cum.
0	21	24.71	24.71
1	64	75.29	100.00
Total	85	100.00	

Regression Analysis

```
. regress booksissued latereturn pref_books libcards

      Source |       SS       df       MS              Number of obs =      85
-------------+------------------------------           F(  3,    81) =    6.79
       Model |  75.275638     3  25.0918793           Prob > F      =  0.0004
    Residual | 299.500833    81  3.69754114           R-squared     =  0.2009
-------------+------------------------------           Adj R-squared =  0.1713
       Total | 374.776471    84  4.46162465           Root MSE      =  1.9229

------------------------------------------------------------------------------
 booksissued |      Coef.   Std. Err.      t    P>|t|     [95% Conf. Interval]
-------------+----------------------------------------------------------------
  latereturn |  -.0130062   .0139947    -0.93   0.355    -.0408512    .0148388
  pref_books |   .6399894   .4847463     1.32   0.190    -.3245037    1.604482
    libcards |   .6738886    .156825     4.30   0.000      .361856    .9859212
       _cons |   .0783467   .6557622     0.12   0.905    -1.226414    1.383108
------------------------------------------------------------------------------
```

1. After running the regression we found a negative coefficient for latereturn. This shows that a unit increase in the number of days delayed in returning a book (latereturn), would decrease the number of books issued (booksissued) by 0.0130062.
2. The positive coefficient for the preference of books (pref_books) and library cards (libcards) shows that an increase in any of these factors would increase the number of books issued during a semester. Specifically, a unit increase in library cards would increase the number of books issued by 0.6738886.
3. The error term is measured by _cons and affects the dependent variable by 0.783467. The R-squared value of 0.2009 shows that the independent variables effect the dependent variables.

4. The "F value" and "Prob(F)" statistics test the overall significance of the regression model. Specifically, they test the null hypothesis that all of the regression coefficients are equal to zero. The F-value of 0.0004 shows that the independent variable has an effect on dependent variable and hence the regression is **significant**.
5. The P>|t| values of 0.355 and 0.190 for latereturn and pref_books respectively, is high, indicating that these variables are statistically **insignificant** at the 0.05 level.
6. The P>|t| value of 0.000 for libcards shows that this variable is very **significant** at the 0.05 level. Indicating that number of library cards that a student has greatly affects the number of books issued in a semester.

Regression Diagnostics

Since our regression result gives us two insignificant variables: pref_books and latereturn, we ran diagnostic tests to check for heteroskedasticity (as our variables have large standard errors), correlation and multicollinearity.

We chose the White test and Breusch Pagen test to check for heteroskedasticity. Both test the null hypothesis that the variance of the residuals is homogenous. If the p-value is very small, we would reject the null hypothesis and accept the alternative hypothesis that the variance is not homogenous.

1. **White test:** P-value = 0.65

```
Cameron & Trivedi's decomposition of IM-test
```

Source	chi2	df	p
Heteroskedasticity	1.90	4	0.7533
Skewness	2.10	2	0.3498
Kurtosis	1.09	1	0.2963
Total	5.10	7	0.6482

2. **Breusch Pagen Test**: P-value = 0.76

```
. estat hettest

Breusch-Pagan / Cook-Weisberg test for heteroskedasticity
         Ho: Constant variance
         Variables: fitted values of booksissued

         chi2(1)      =     0.09
         Prob > chi2  =   0.7649
```

Given these high p-values, we can comfortably reject the alternative hypothesis that the variance is not homogenous and the model does not suffer from heteroskedasticity.

3. **Correlation**: checking if the independent variables are highly correlated.

```
. correlate booksissued latereturn pref_books libcards
(obs=85)
```

	booksi~d	latere~n	pref_b~s	libcards
booksissued	1.0000			
latereturn	-0.0299	1.0000		
pref_books	0.1310	0.0692	1.0000	
libcards	0.4203	0.1261	0.0139	1.0000

Our result shows that there is not a significant correlation among our independent variables and this does not contribute to the statistically insignificant independent variables.

As the degree of Multicollinearity increases, the estimates of the coefficients of the independent variables become unstable and the standard errors get inflated. We used the **vif** (Variation Inflation Factor) after we ran the regression to check for multicollinearity. A variance whose VIF values are greater than 10 is generally a cause of concern and the tolerance level (1/ VIF) checks on the degree of collinearity. A tolerance level lower than 0.1 merits for further investigation.

Mean VIF: 1.01, 1/VIF > 0.97 for all independent variables.

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. regress booksissued latereturn pref_books libcards

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       _cons |   .0783467   .6557622     0.12   0.905    -1.226414    1.383108
------------------------------------------------------------------------------

. vif

    Variable |       VIF       1/VIF
-------------+----------------------
  latereturn |      1.02    0.979547
    libcards |      1.02    0.984070
  pref_books |      1.00    0.995185
-------------+----------------------
    Mean VIF |      1.01
```

We can now conclude that our regression does not suffer from multicollinearity

Final Remarks

After running the regression diagnostics, we can confidently confirm that our model doesn't suffer from multicollinearity or heteroscedasticity. The two statistically insignificant variables could be because of some endogeneity in the model i.e. the error term is correlated with one or more of the independent variables. We could use Instrumental Variables to fix this problem for future research.

The R-squared value of 0.20 that is not very high could be because we omitted a significant variable, which is currently a part of the error term. These results from the project can be corrected in future research.

Our final result is that there is not a significant relationship between the preference of books over PDFs to the number of books issued. This confirms our second result that we may have excluded a significant variable, which is a part of the error term.

Perhaps a better tool to check if the people who prefer books actually issue them is a probit model. However this is beyond the scope of our course.

Appendix - 1

Survey

Your email address (**pt705@snu.edu.in**) will be recorded when you submit this form. Not you? Sign out

Do you usually issue books from the library?

- ◯ Yes
- ◯ No

On average, how may books do you issue in a semester?

Your answer

By how many days are you usually late in returning a library book?

Your answer

Do you prefer using books to pdfs/ppt?

- ◯ Yes
- ◯ No

How many library cards do you have?

Your answer